



UNIVERSITÄT
LEIPZIG

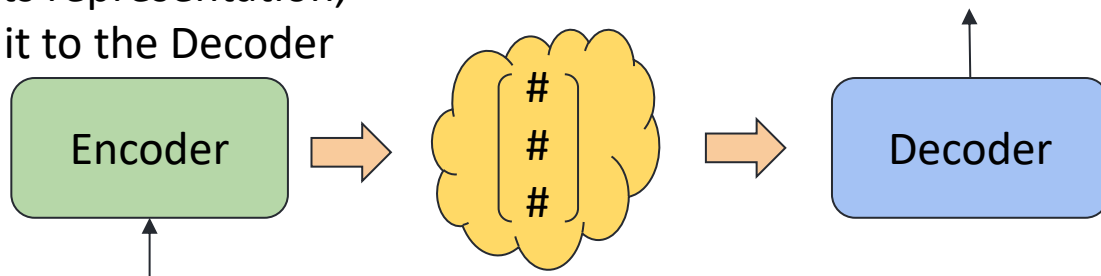
LLMs 1: BERT

Natural Language Processing

Leonie Weissweiler

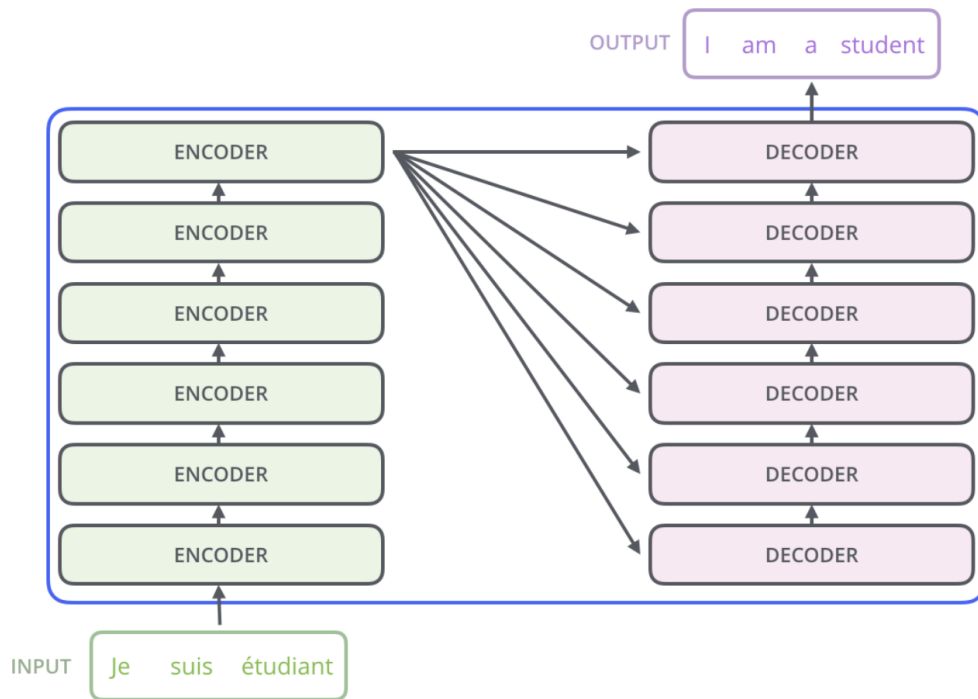
RECAP

Encoder reads the *source*,
produces its representation,
and gives it to the Decoder

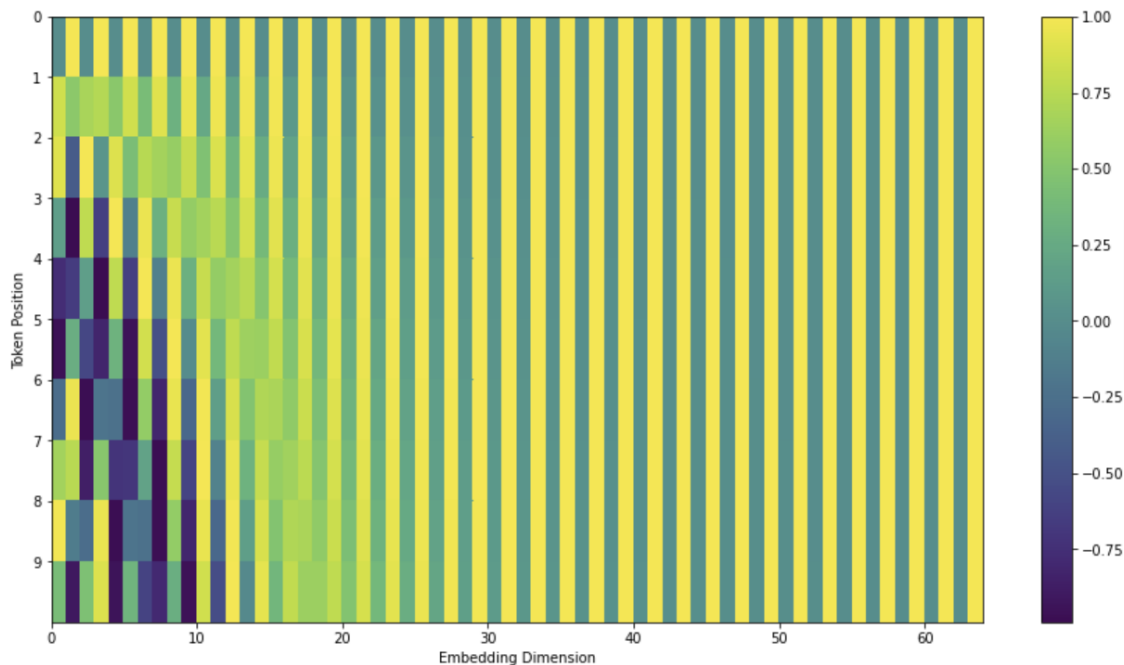


Decoder uses the
source representation
to generate the *output*

ENCODER AND DECODER IN BLOCKS



GENERATING POSITIONAL ENCODINGS



$$\text{PE}_{pos,2i} = \sin(pos/10000^{2i/d_{model}}),$$

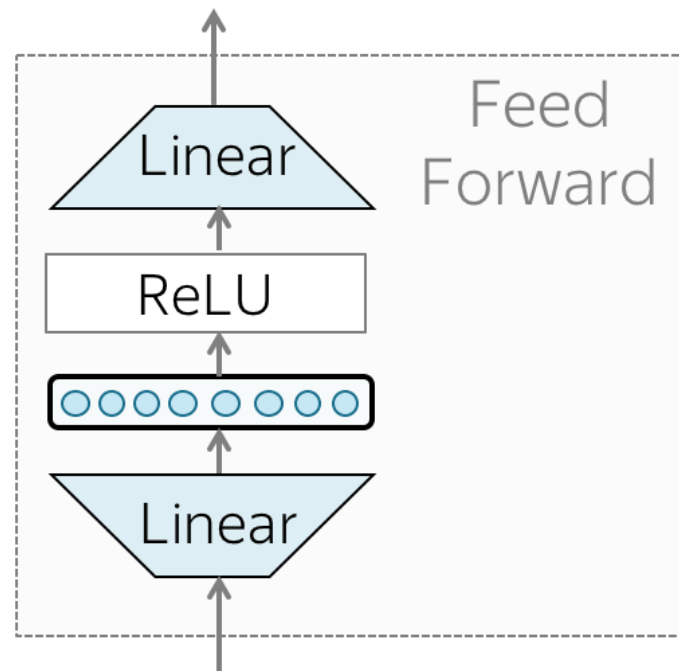
$$\text{PE}_{pos,2i+1} = \cos(pos/10000^{2i/d_{model}}),$$

FEED-FORWARD BLOCKS

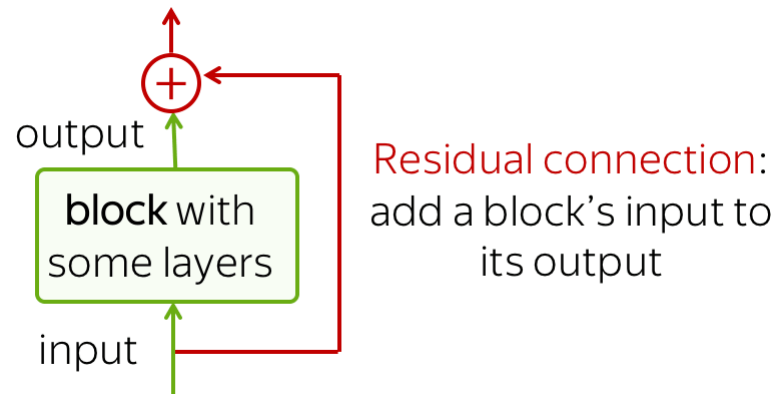
- In addition to attention, each layer has a feed-forward network block: two linear layers with ReLU non-linearity between them:

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2.$$

- After looking at other tokens via an attention mechanism, a model uses an FFN block to process this new information (attention - "look at other tokens and gather information", FFN - "take a moment to think and process this information").



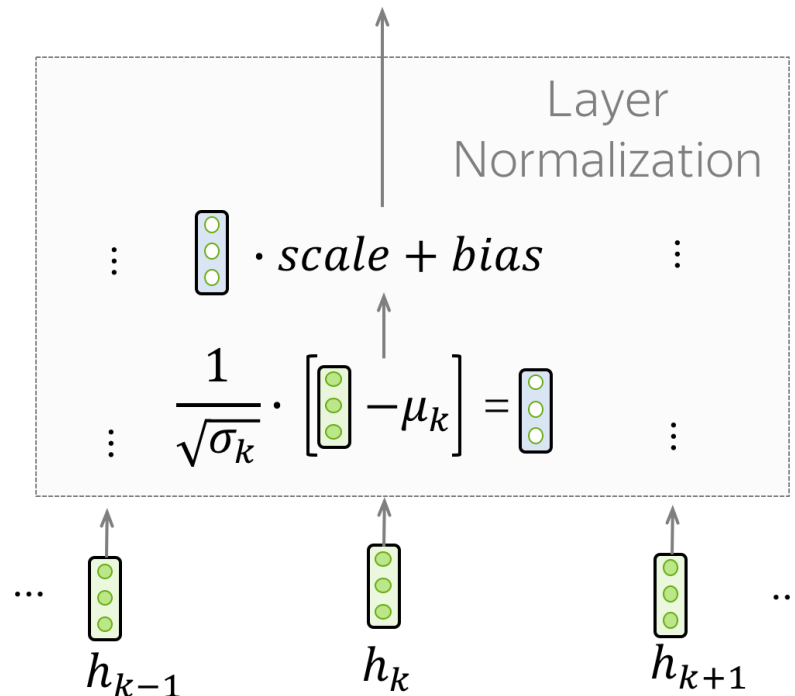
RESIDUAL CONNECTIONS



- Residual connections are very simple (add a block's input to its output), but at the same time are very useful: they ease the gradient flow through a network and allow stacking a lot of layers.
- In the Transformer, residual connections are used after each attention and FFN block. On the illustration above, residuals are shown as arrows coming around a block to the green "Add & Norm" layer. In the "Add & Norm" part, the "Add" part stands for the residual connection.

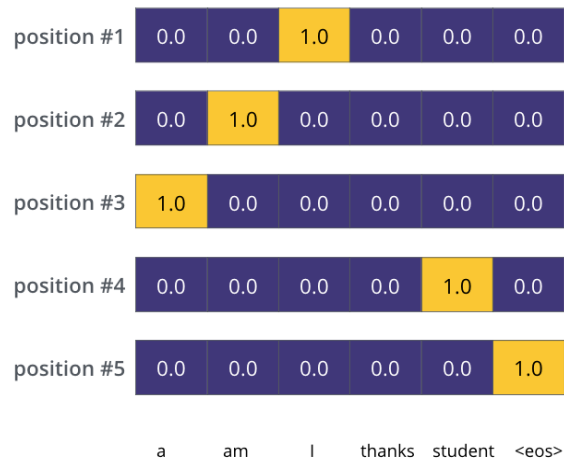
LAYER NORMALISATION

- independently normalizes vector representation of each example in batch - this is done to control "flow" to the next layer
- improves convergence stability and sometimes even quality
- LayerNorm has trainable parameters, scale and bias, which are used after normalization to rescale layer's outputs (or the next layer's inputs)
- Scale and bias are trainable, but they are layer parameters $\rightarrow$ the same for every example



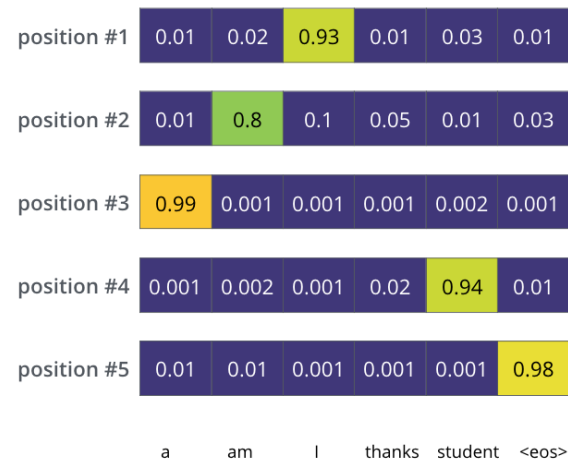
Target Model Outputs

Output Vocabulary: a am I thanks student <eos>

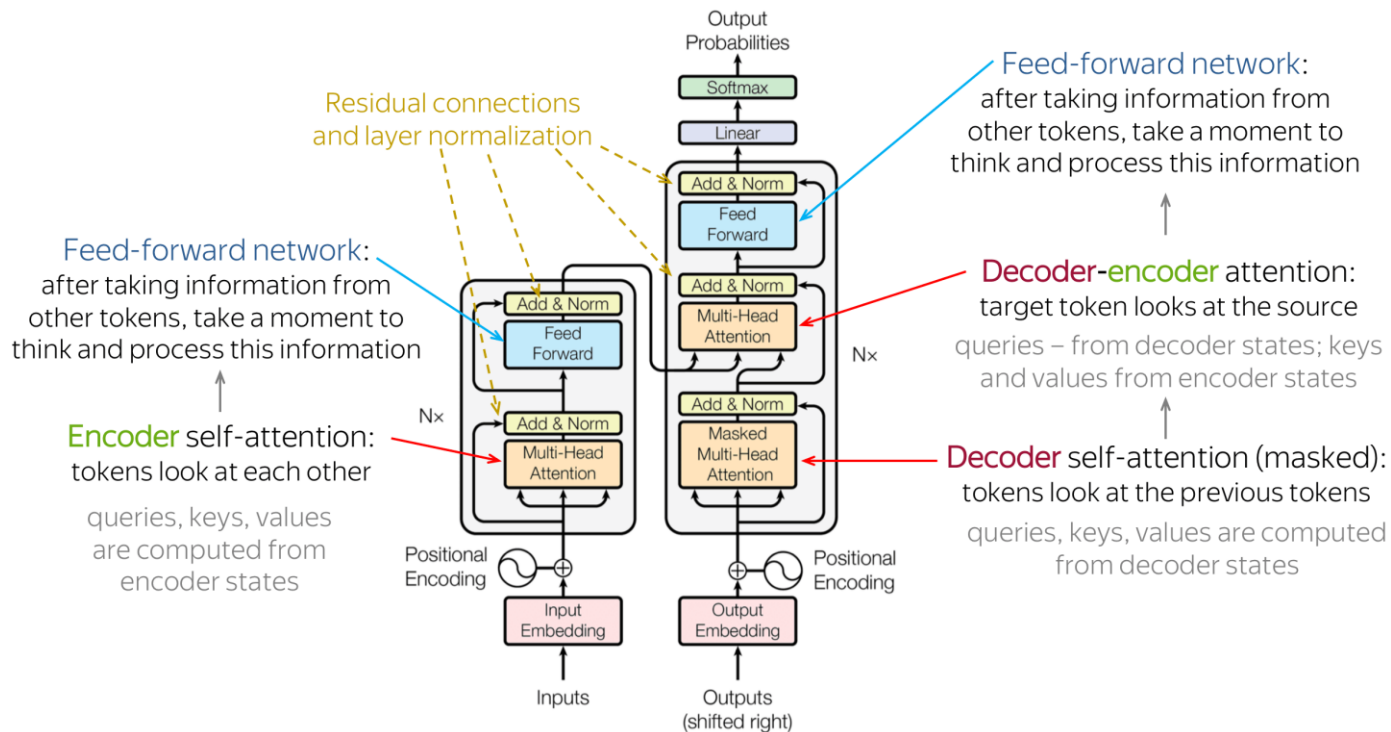


Trained Model Outputs

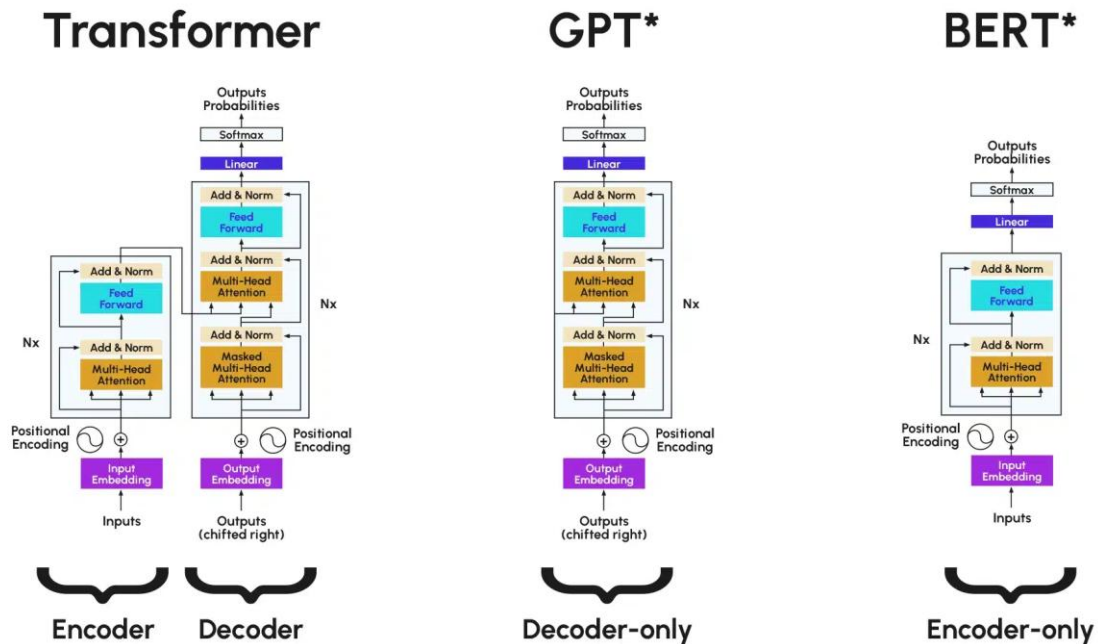
Output Vocabulary: a am I thanks student <eos>



PUTTING IT ALL TOGETHER



PREVIEW: ENCODER-ONLY AND DECODER-ONLY



*Illustrative example, exact model architecture may vary slightly

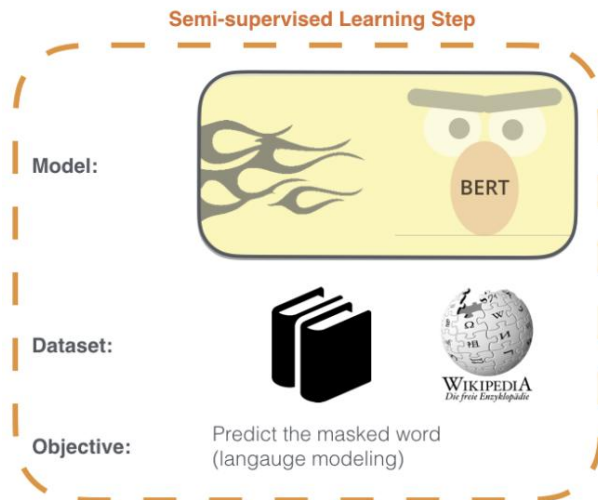
QUESTIONS?

PRETRAINING AND FINETUNING

BERT

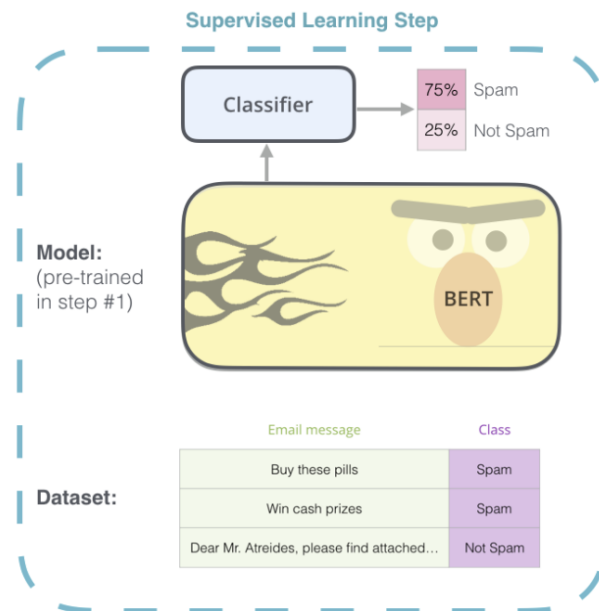
1 - **Semi-supervised** training on large amounts of text (books, wikipedia..etc).

The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.



Pretraining: train the language model to predict the masked word, on large amounts of unlabelled text

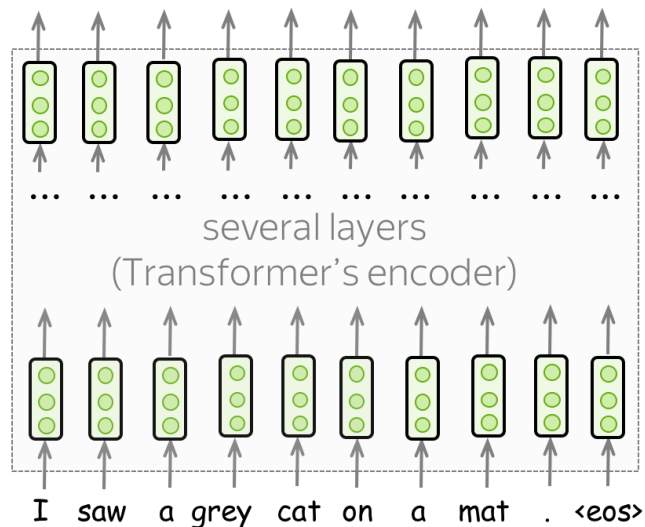
2 - **Supervised** training on a specific task with a labeled dataset.



Finetuning: put a classification head (essentially a small FFN with an output layer of appropriate size) on top of the pretrained model, then finetune the entire thing with the labelled data for a specific task

MASKED LANGUAGE MODELLING

BERT: BIDIRECTIONAL ENCODER REPRESENTATIONS FROM TRANSFORMERS



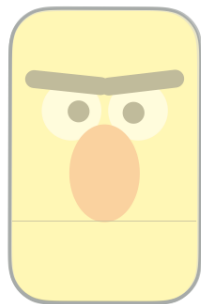
Model architecture:

- Transformer's encoder

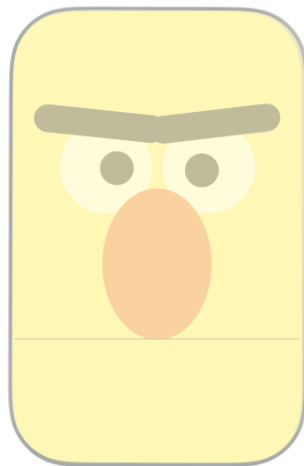
What is special about it:

- Training objectives
 - MLM: Masked language modeling
 - NSP: Next sentence prediction
- The way it is used
 - No task-specific models

BERT ARCHITECTURE

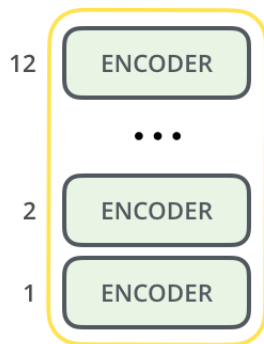


BERT_{BASE}



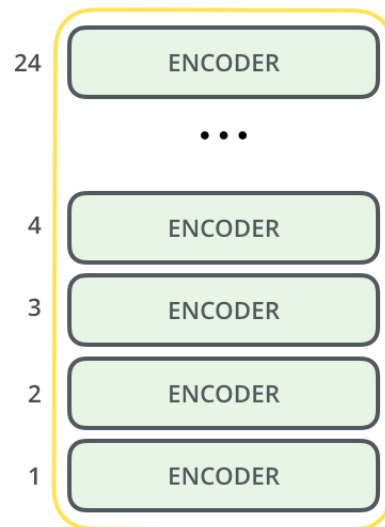
BERT_{LARGE}

BERT ARCHITECTURE



BERT_{BASE}

- 12 encoder layers
- 768 hidden units
- 12 attention heads



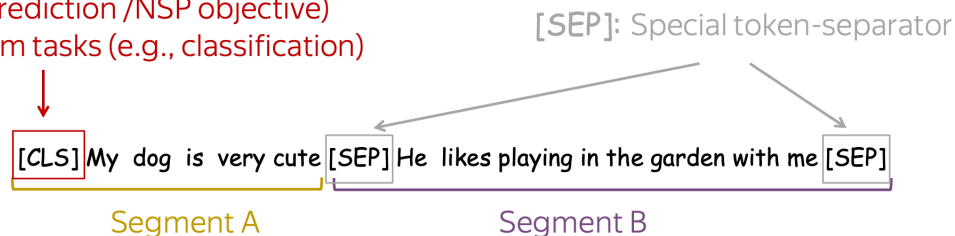
BERT_{LARGE}

- 16 encoder layers
- 1024 hidden units
- 16 attention heads

TRAINING INPUT: PAIRS OF SENTENCES WITH SPECIAL TOKENS

[CLS]: Special token

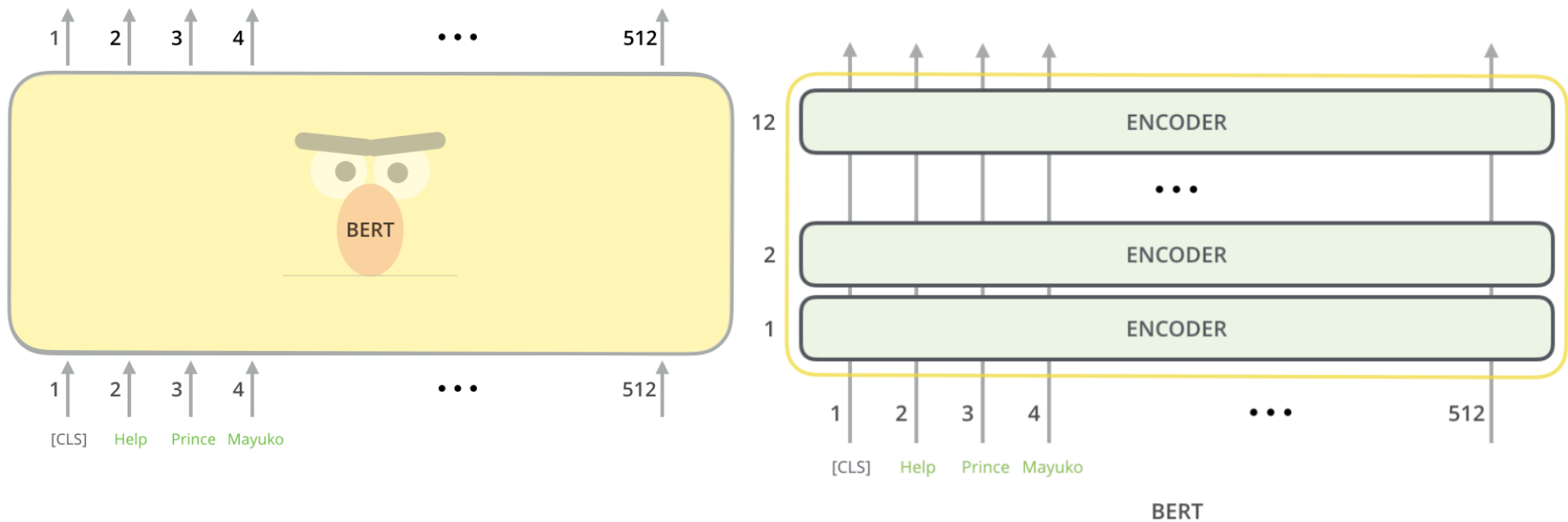
- Training time: predict if sentences are consecutive or not (Next Sentence Prediction /NSP objective)
- Test time: downstream tasks (e.g., classification)



Training on pairs of sentences: either consecutive or random (50%/50%)

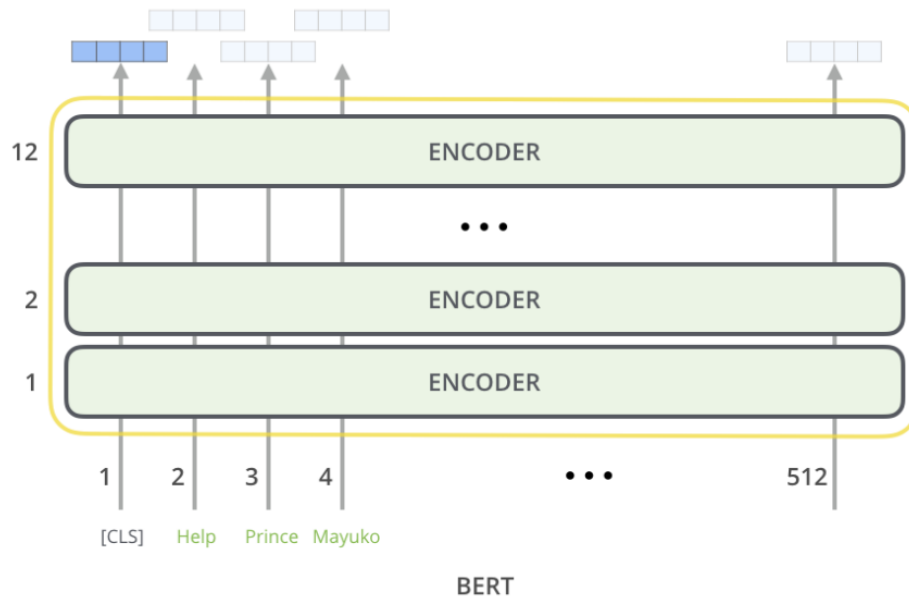
- In training, BERT sees pairs of sentences separated with a special token-separator [SEP]
- Another special token is [CLS].
 - In training, it is used for the NSP objective
 - Once a model is trained, it is used for downstream tasks.

BERT INPUT



[CLS] is the special classification token

BERT OUTPUT



One vector of hidden size (e.g. 768 for BERT Base) for each input token

QUESTIONS?

HOW TO PRE-TRAIN BERT

TWO PRETRAINING OBJECTIVES: MLM AND NSP

Training time: predict if sentences
are consecutive (NSP objective)

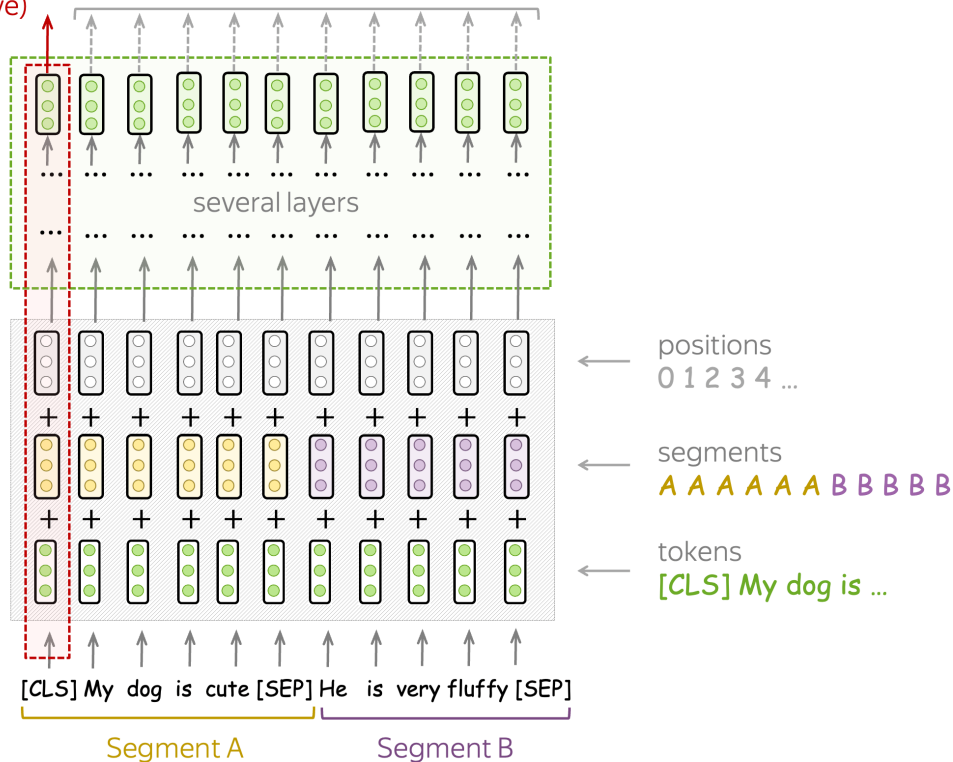
Test time: classification

Training time: MLM objective

Model
(Transformer
encoder)

Input

Training on pairs of
sentences: either
consecutive or
random (50%/50%)

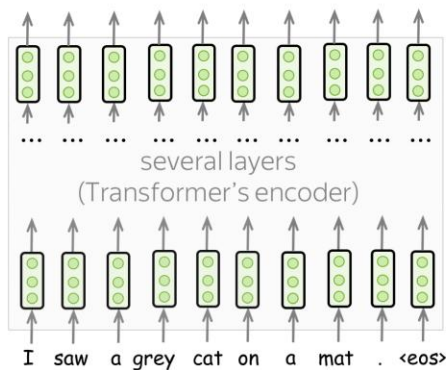


MASKED LANGUAGE MODELLING

At each training step:

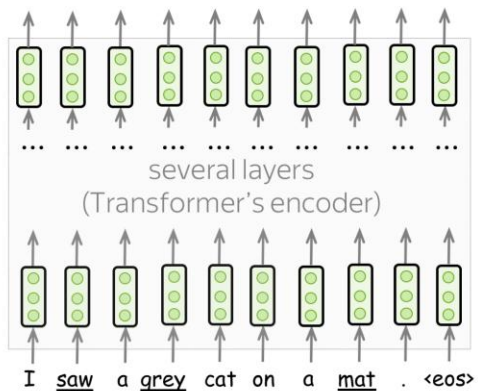
- select some tokens (each token is selected with the probability of 15%)
- replace these selected tokens
 - with the special token [MASK] with $p=80\%$,
 - with a random token with $p=10\%$,
 - with the original token (remain unchanged) with $p=10\%$
- predict original tokens (compute loss)

MASKED LANGUAGE MODELLING



At each training step:

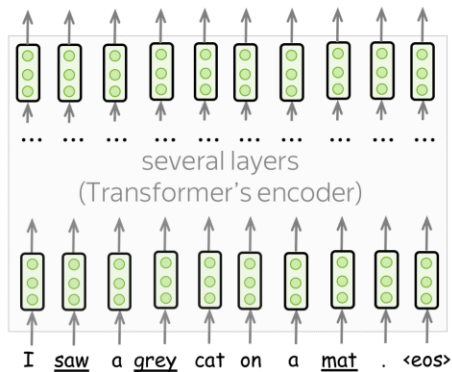
MASKED LANGUAGE MODELLING



At each training step:

- pick randomly 15% of tokens

MASKED LANGUAGE MODELLING

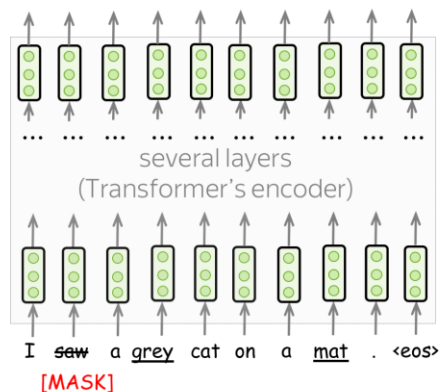


At each training step:

- pick randomly 15% of tokens
- replace each of the chosen tokens with something



MASKED LANGUAGE MODELLING



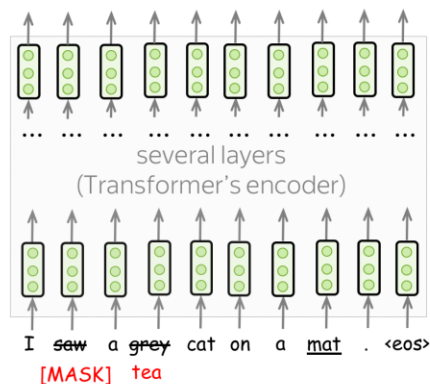
- [MASK],
with $p = 80\%$

At each training step:

- pick randomly 15% of tokens
- replace each of the chosen tokens with something



MASKED LANGUAGE MODELLING

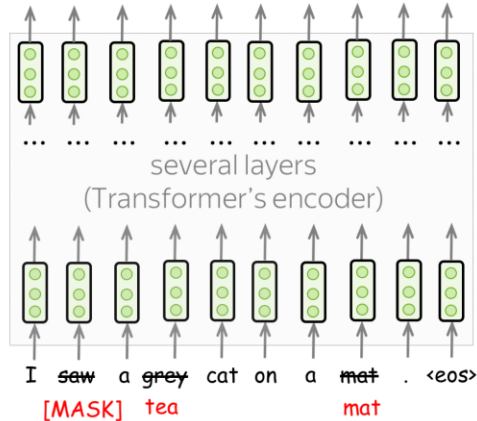


At each training step:

- pick randomly 15% of tokens
- replace each of the chosen tokens with something

- [MASK], with $p = 80\%$
- Random token, with $p = 10\%$

MASKED LANGUAGE MODELLING

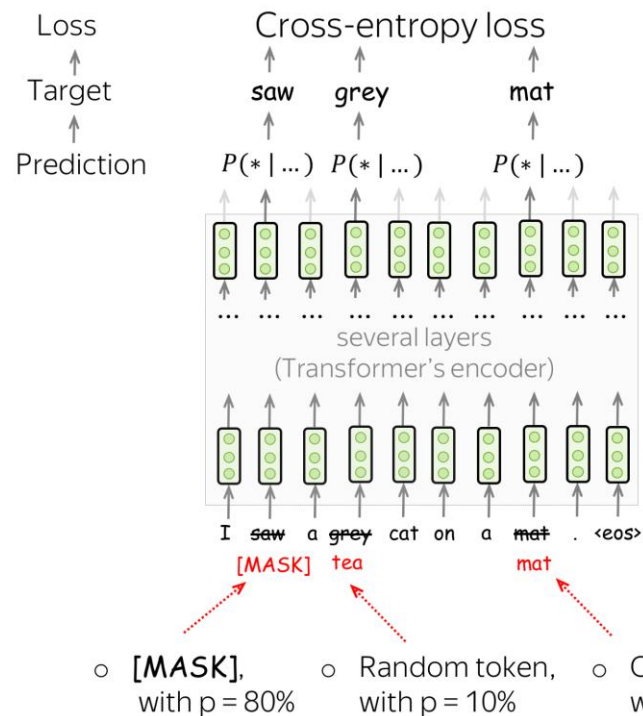


At each training step:

- pick randomly 15% of tokens
- replace each of the chosen tokens with something

- [MASK], with $p = 80\%$
- Random token, with $p = 10\%$
- Original token, with $p = 10\%$

MASKED LANGUAGE MODELLING

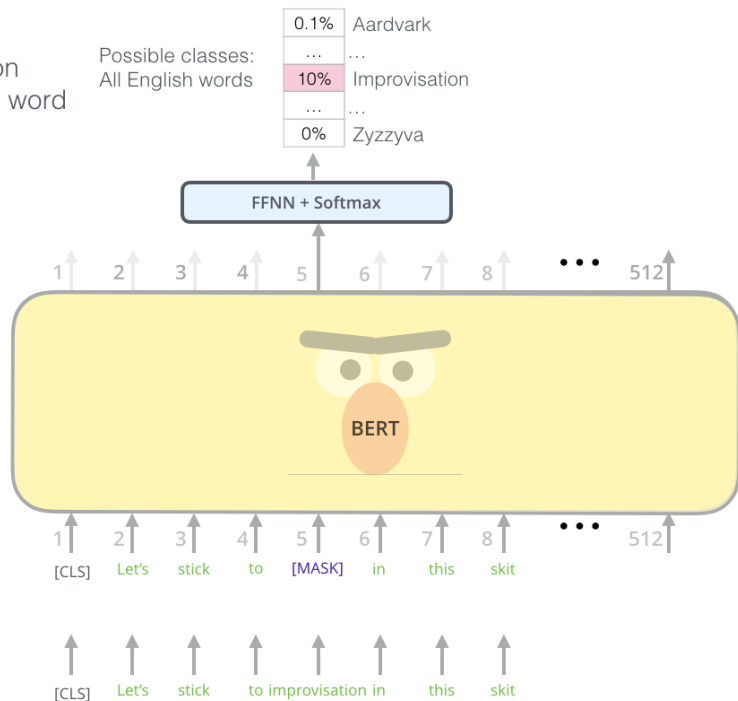


At each training step:

- pick randomly 15% of tokens
- replace each of the chosen tokens with something
- predict original chosen tokens

MASKED LANGUAGE MODELLING

Use the output of the masked word's position to predict the masked word



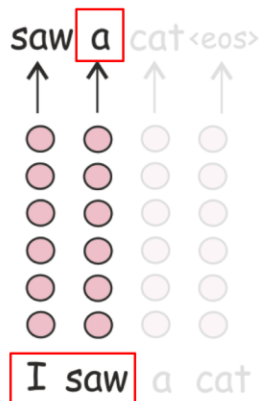
Randomly mask
15% of tokens

Input

MLM VS. AUTOREGRESSIVE LANGUAGE MODELLING

Language Modeling

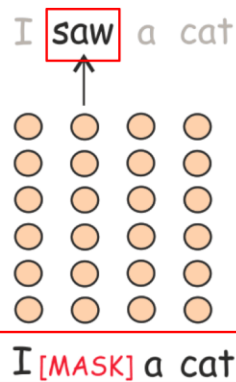
- Target: next token
- Prediction: $P(* | \text{I saw})$



left-to-right, does
not see future

Masked Language Modeling

- Target: current token (the true one)
- Prediction: $P(* | \text{I [MASK] a cat})$



sees the whole text, but
something is corrupted

WHY, AND WHY THESE PERCENTAGES?

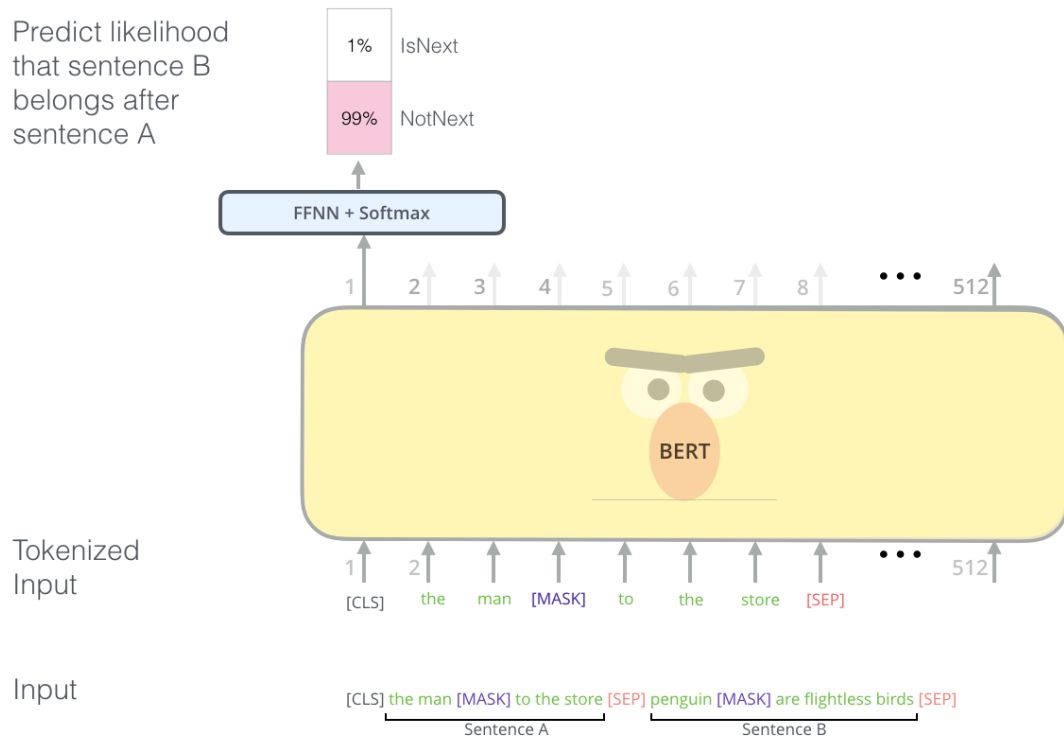
- If we don't mask the tokens we later predict, the model can easily recover the information by attending 100% to the left in layer one and then 100% to the right in layer two
- If we only predict from masked tokens, there would be a mismatch between pre-training and fine-tuning because the [MASK] token will never appear during fine-tuning
- The rest is magic!

QUESTIONS?

NEXT SENTENCE PREDICTION

- Binary classification task
- From the final-layer representation of the special token [CLS], the model predicts whether the two sentences are consecutive sentences in some text or not
- in training, 50% of examples contain consecutive sentences extracted from training texts and another 50% - a random pair of sentences
- Input: [CLS] the man went to [MASK] store [SEP] he bought a gallon [MASK] milk [SEP]
Label: isNext
- Input: [CLS] the man went to [MASK] store [SEP] penguin [MASK] are flight ##less birds [SEP]
Label: notNext

NEXT SENTENCE PREDICTION



Idea: make BERT better at handling relationships between multiple sentences

HOW TO PRETRAIN BERT: DETAILS AND DATA

- Data
 - BooksCorpus (800M words)
 - English Wikipedia (2500M words)
- Batch size of 256 sequences → perform one gradient update for them together
- 40 epochs → go over the entire data 40 times

TWO PRETRAINING OBJECTIVES: MLM AND NSP

Training time: predict if sentences
are consecutive (NSP objective)

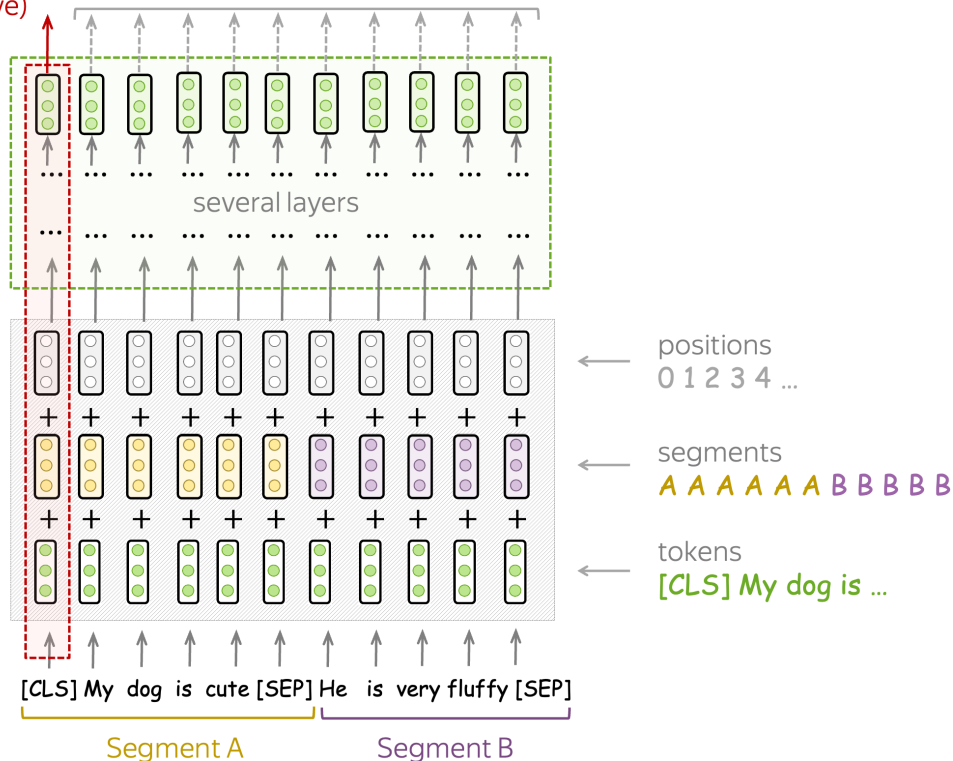
Test time: classification

Training time: MLM objective

Model
(Transformer
encoder)

Input

Training on pairs of
sentences: either
consecutive or
random (50%/50%)



QUESTIONS?

BERT CONTEXTUAL EMBEDDINGS AFTER PRETRAINING

STATIC VS CONTEXTUAL EMBEDDINGS

- Static embeddings (e.g. word2vec) represent word types (dictionary entries)
- Contextual embeddings represent word instances (one for each time the word occurs in any context/sentence)

German article “die”

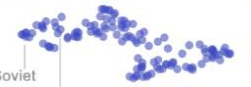


Was der Fall ist, **die** Tatsache, ist das Bestehen von Sachverhalten.

über **die** Verhandlungen der Königl.

This figure shows a scatter plot of blue dots representing contextual embeddings for the word "die" in German. Two lines point from the plot to two example sentences. In the first sentence, "die" is highlighted in red. In the second sentence, "die" is also highlighted in red.

single person dies ← → multiple people die



Chernenko became the first Soviet leader to **die** in less than three years

Vaughan's ultimate fantasy was to **die** in a head-on collision with movie star Elizabeth Taylor

Over 60 people **die** and over 100 are unaccounted for.

Many more **die** from radiation sickness, starvation and cold.

This figure shows a scatter plot of blue dots representing contextual embeddings for the word "die" in English. A double-headed arrow above the plot indicates a spectrum from "single person dies" to "multiple people die". Four lines point from the plot to four example sentences. In each sentence, "die" is highlighted in red.

a playing die



Players must always move a token according to the **die** value

The faces of a **die** may be placed clockwise or counterclockwise

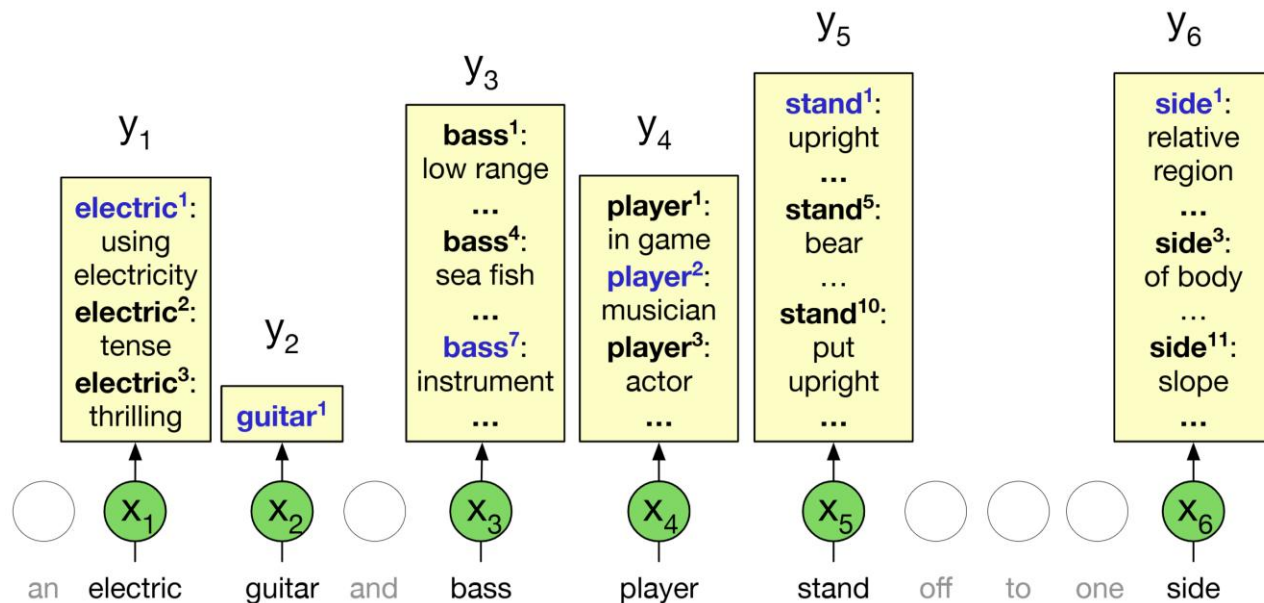
This figure shows a scatter plot of blue dots representing contextual embeddings for the word "die" in English. Two lines point from the plot to two example sentences. In the first sentence, "die" is highlighted in red. In the second sentence, "die" is also highlighted in red.

WORD SENSES

- Words are ambiguous
- A word sense is a discrete representation of one aspect of meaning
 - Mouse₁: ... a mouse controlling a computer system in 1968
 - Mouse₂: ... a quiet animal like a mouse
 - Bank₁: ... a bank can hold the investments in a custodial account ...
 - Bank₂: ... as agriculture burgeons on the east bank, the river ...
- Contextual embeddings offer a continuous high-dimension model of meaning that is more fine-grained than discrete senses

WORD SENSE DISAMBIGUATION

The task of selecting the correct sense for a word



1-NEAREST NEIGHBOUR ALGORITHM FOR WSD

Training

- Take a sense-labelled corpus, e.g. SEMCOR
- Run the corpus through BERT to get contextual embeddings for each token
- Then for each sense s of a word w for n tokens of that sense, pool the embeddings:

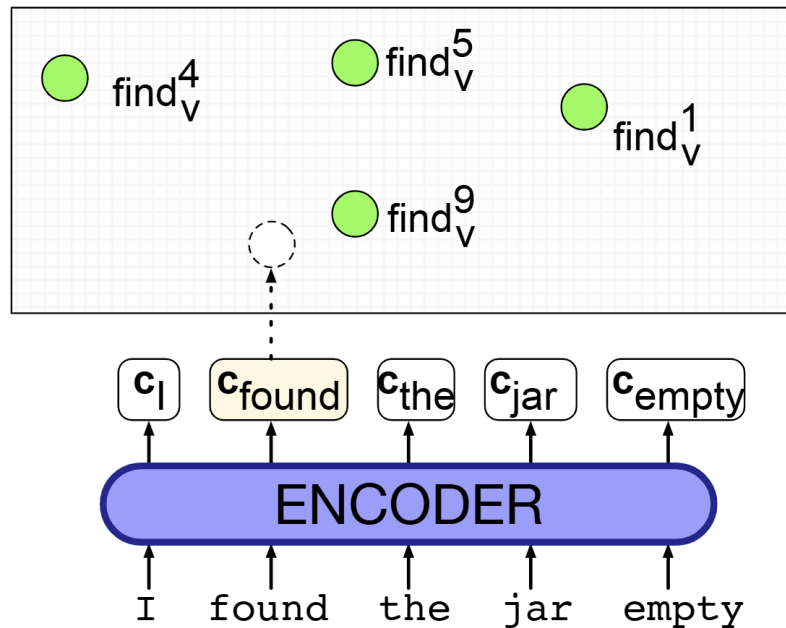
$$\mathbf{v}_s = \frac{1}{n} \sum_i \mathbf{v}_i \quad \forall \mathbf{v}_i \in \text{tokens}(s)$$

Testing

Given a token of a target word t , compute the contextual embedding of t and choose its nearest neighbour sense from the training set

$$\text{sense}(t) = \underset{s \in \text{senses}(t)}{\operatorname{argmax}} \operatorname{cosine}(\mathbf{t}, \mathbf{v}_s)$$

1-NEAREST NEIGHBOUR ALGORITHM FOR WSD

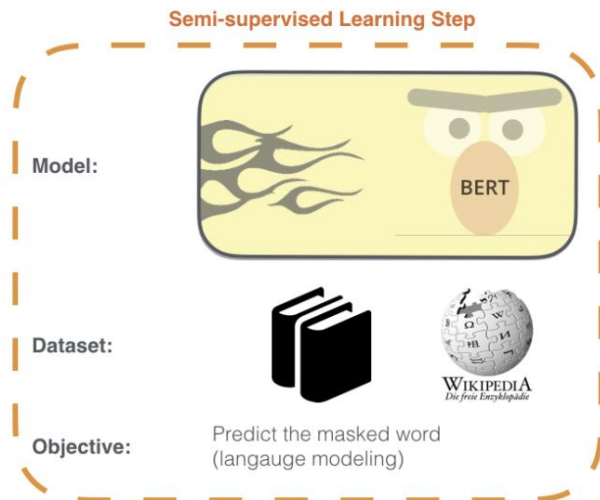


QUESTIONS?

TRANSFER LEARNING

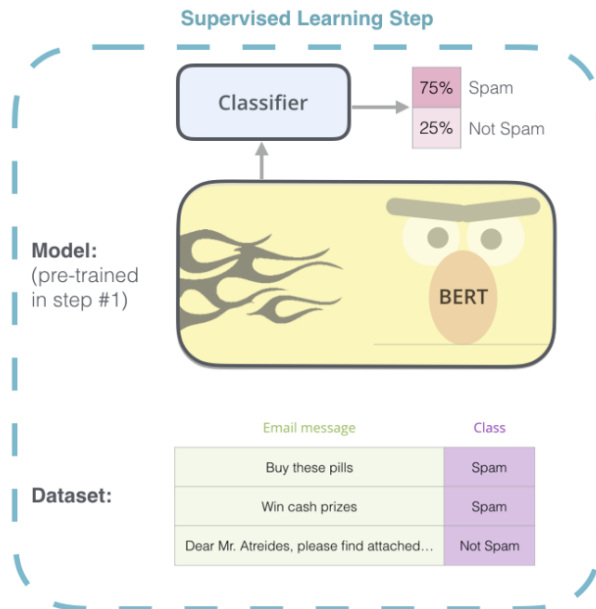
1 - Semi-supervised training on large amounts of text (books, wikipedia..etc).

The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.



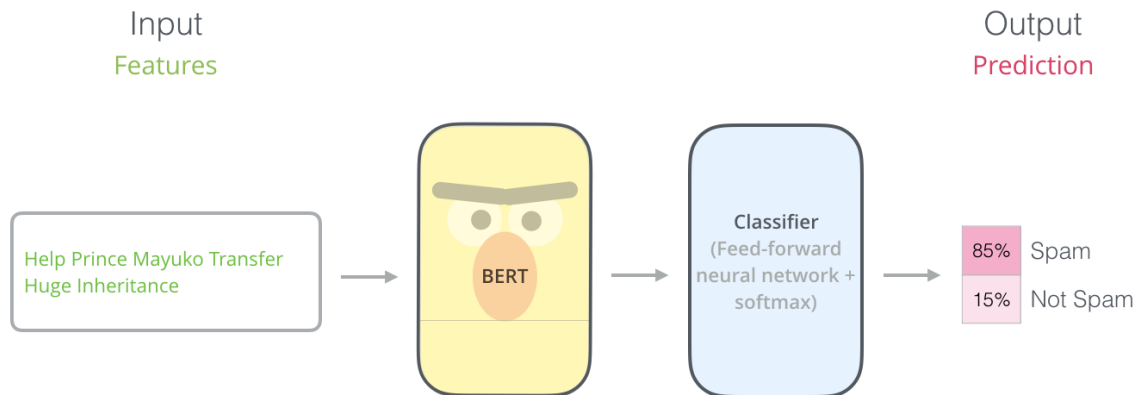
Pretraining: train the language model to predict the masked word, on large amounts of unlabelled text

2 - Supervised training on a specific task with a labeled dataset.



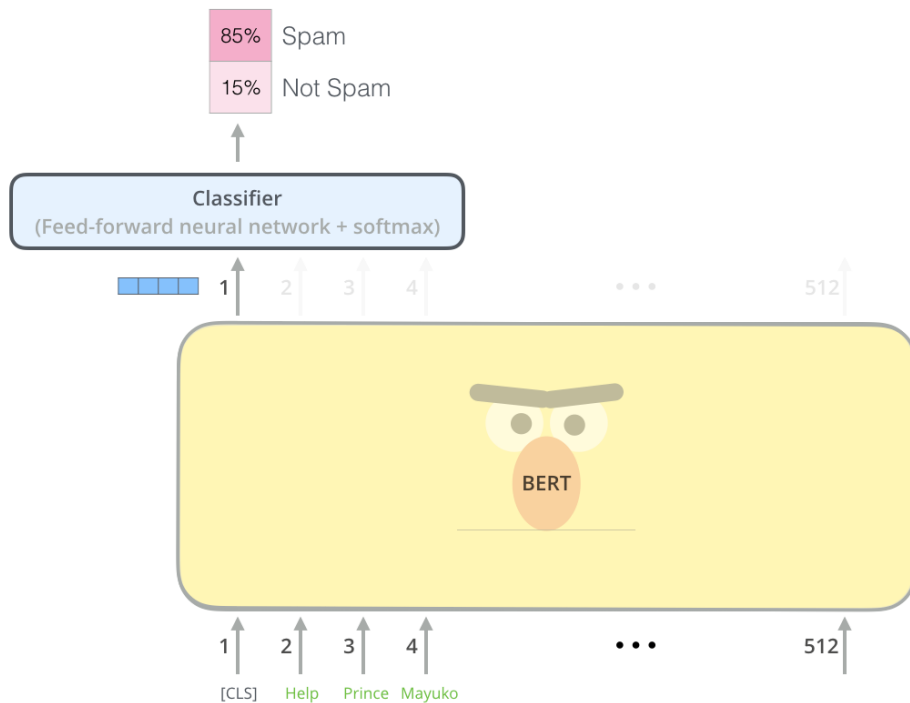
Finetuning: put a classification head (essentially a small FFN with an output layer of appropriate size) on top of the pretrained model, then finetune the entire thing with the labelled data for a specific task

SENTENCE CLASSIFICATION WITH BERT

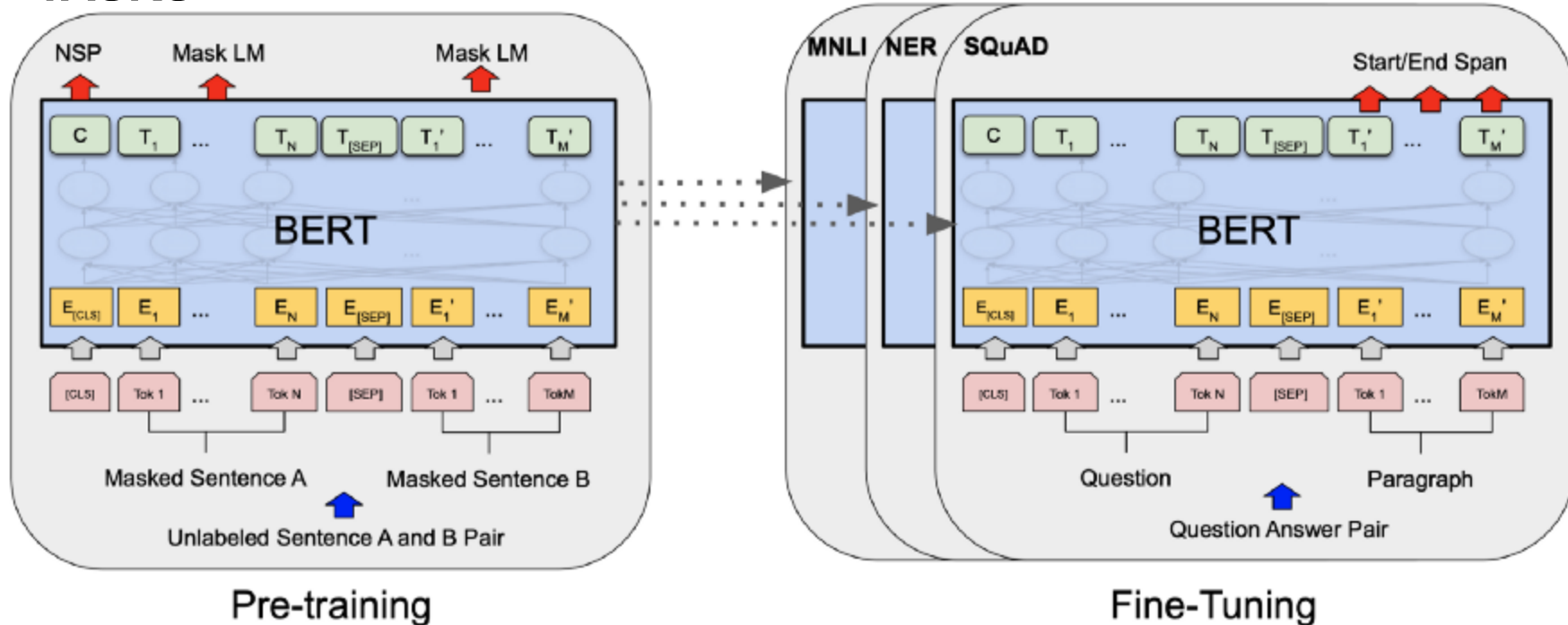


Email message	Class
Buy these pills	Spam
Win cash prizes	Spam
Dear Mr. Atreides, please find attached...	Not Spam

CLASSIFICATION WITH BERT



TRANSFER LEARNING: FINE-TUNING BERT ON DIFFERENT TASKS



IMPORTANT TERMINOLOGY: HEAD

- Head: a small FFN placed on top of the encoder model somewhere
- Input dimensionality = hidden layer size of the encoder model
- Output dimensionality: appropriate to the task
 - Language modelling head: output dimensionality = vocabulary size
 - Binary classification head: output dimensionality = 1
 - Labelling head: output dimensionality = number of labels

SINGLE SENTENCE CLASSIFICATION

Assign a single label to the entire sentence

- Linguistic acceptability
- Sentiment classification

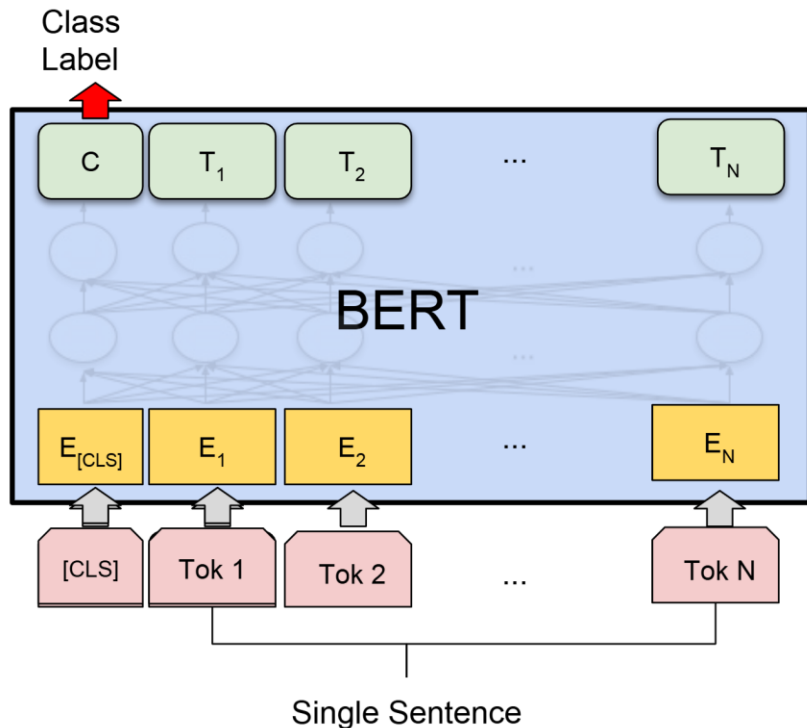
LINGUISTIC ACCEPTABILITY JUDGEMENTS

Is a given sentence grammatical in the English language?

✓ = acceptable, * = unacceptable

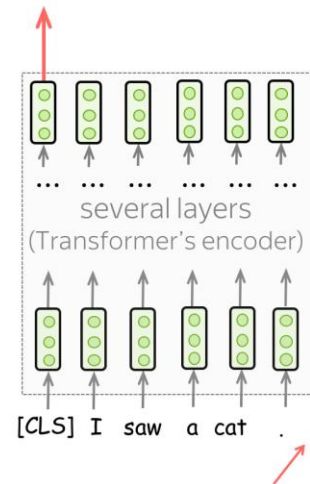
Label	Sentence
*	The more books I ask to whom he will give, the more he reads.
✓	I said that my father, he was tight as a hoot-owl.
✓	The jeweller inscribed the ring with the name.
*	many evidence was provided.
✓	They can sing.
✓	The men would have been all working.
*	Who do you think that will question Seamus first?
*	Usually, any lion is majestic.
✓	The gardener planted roses in the garden.
✓	I wrote Blair a letter, but I tore it up before I sent it.

SINGLE SENTENCE CLASSIFICATION WITH BERT



- Use the last hidden state of the [CLS] token
- Binary classification head: acceptable or not acceptable

class label



No second sentence!

SENTENCE PAIR CLASSIFICATION

Assign a label to pairs of sentences:

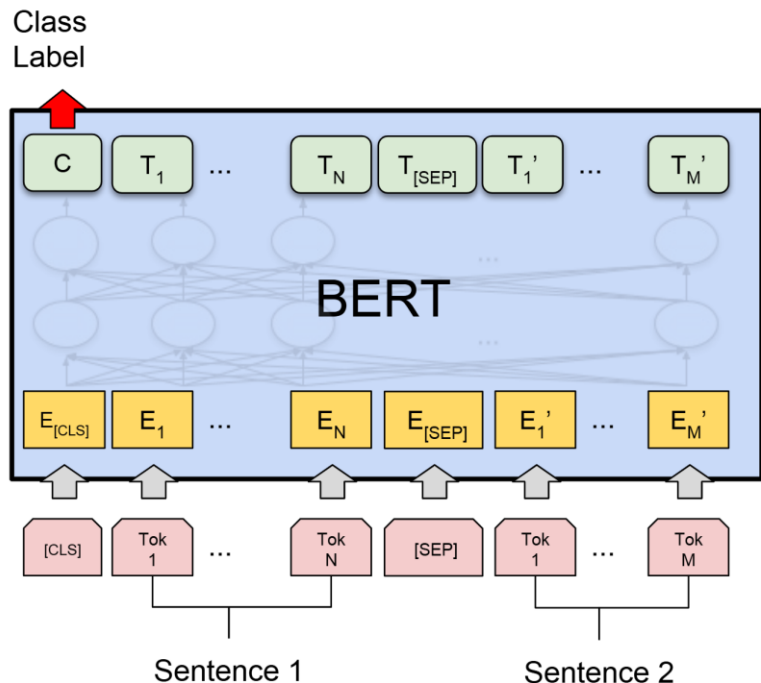
- Paraphrase detection (are the two sentences paraphrases of each other?)
- Logical entailment (does sentence A logically entail sentence B?)
- Discourse coherence (how coherent is sentence B as a follow-on to sentence A?)

EXAMPLE FOR SENTENCE PAIR CLASSIFICATION: NATURAL LANGUAGE INFERENCE

Pairs of sentences are given one of three labels:

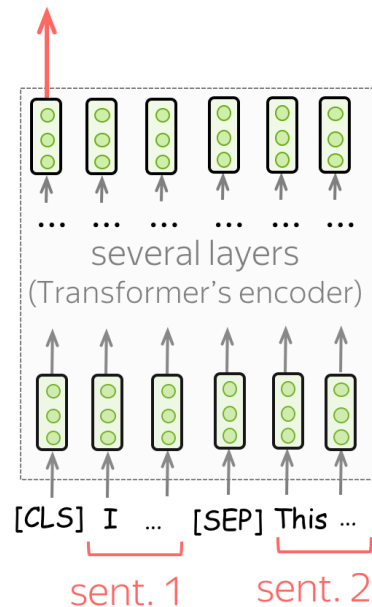
- Neutral
 - A: Jon walked back to the town to the smithy.
 - B: Jon travelled back to his hometown.
- Contradiction
 - A: Tourist Information offices can be very helpful.
 - B: Tourist Information offices are never of any help.
- Entailment
 - A: I'm confused
 - B: Not all of it is very clear to me.

SENTENCE PAIR CLASSIFICATION WITH BERT



- Use the output vector for the [CLS] token as the input to the classification head
- Note: only one classification head for a decision that is about two sentences
- NLI example: 3 labels as output neurons (neutral, contradiction, entailment)

class label



QUESTION ANSWERING

Given a span of text and a question, extract the segment of text that constitutes the answer to the question.

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **grau-pel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called “showers”.

What causes precipitation to fall?

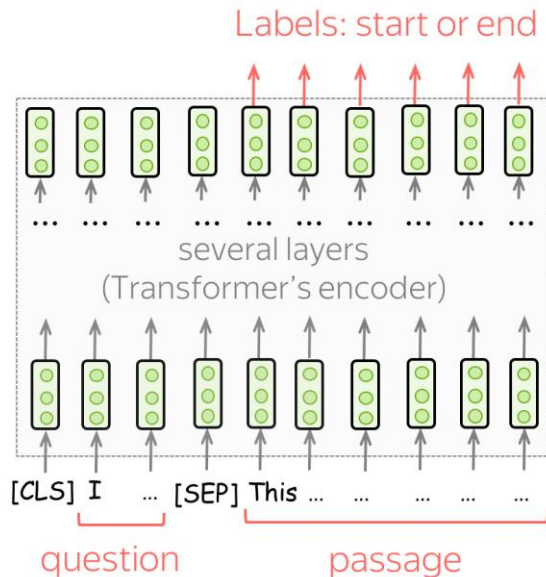
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

grau-pel

Where do water droplets collide with ice crystals to form precipitation?

within a cloud



For each token in the text passage where the answer could be, use the final representations to predict if this token is the start or the end of the correct segment

SINGLE SENTENCE TAGGING / SEQUENCE LABELLING

Assign a label from a small fixed set of labels to each token in the sequence

- Named entity recognition
- Part-of-speech tagging

BIO TAGGING

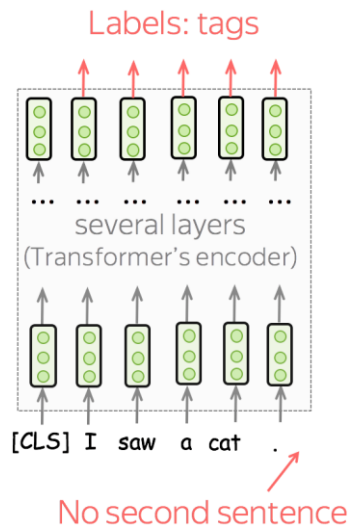
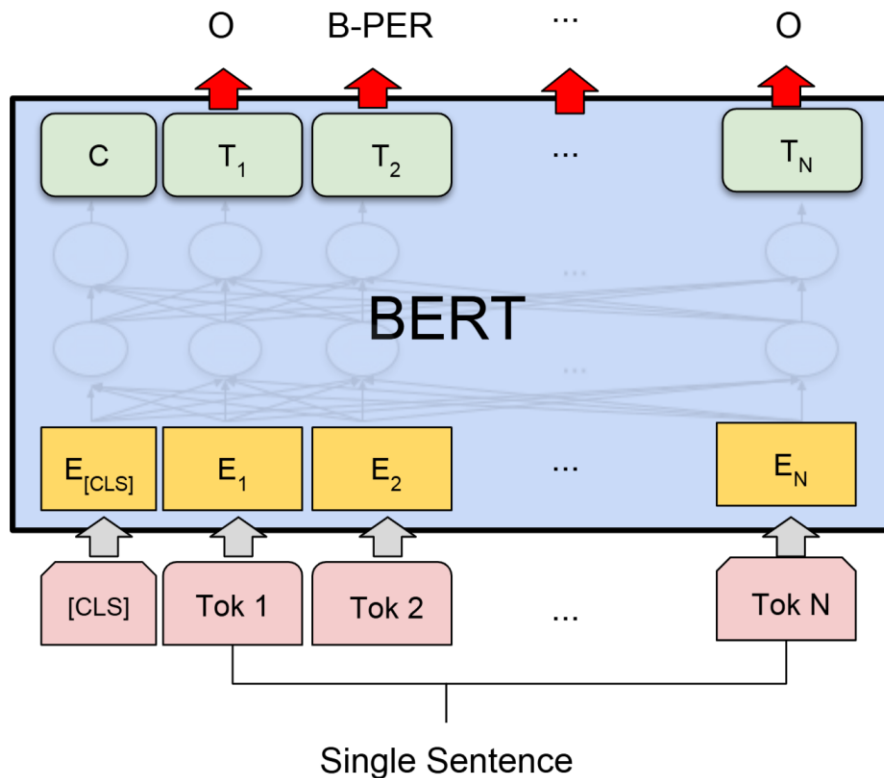
A method that lets us turn a segmentation task (finding boundaries of entities) into a classification task.

E.g. for Named Entity Recognition (NER):

[**PER Jane Villanueva**] of [**ORG United**], a unit of [**ORG United Airlines Holding**], said the fare applies to the [**LOC Chicago**] route.

Words	IO Label	BIO Label
Jane	I-PER	B-PER
Villanueva	I-PER	I-PER
of	O	O
United	I-ORG	B-ORG
Airlines	I-ORG	I-ORG
Holding	I-ORG	I-ORG
discussed	O	O
the	O	O
Chicago	I-LOC	B-LOC
route	O	O
.	O	O

SINGLE SENTENCE TAGGING WITH BERT



- Use one classification head (the same trained network, but with different inputs each time)
- 7-way classification (one beginning, one inside per label, plus one outside)

HOW TO FINETUNE BERT: DETAILS AND DATA

- Data
 - Whatever training data is provided by the datasets of the individual tasks
- 2-4 epochs
- → much faster than pre-training

QUESTIONS?

ROBERTA

ROBERTA

- RoBERTa: Robustly optimised BERT approach
- The best kind of empirical work: re-implemented BERT, tried out which of the components were useful, found improvements for some
- We can find explanations for why each improvement works, but it was really just empirically determined and all explanations are post-hoc
- Built a better model that is now the standard for MLM

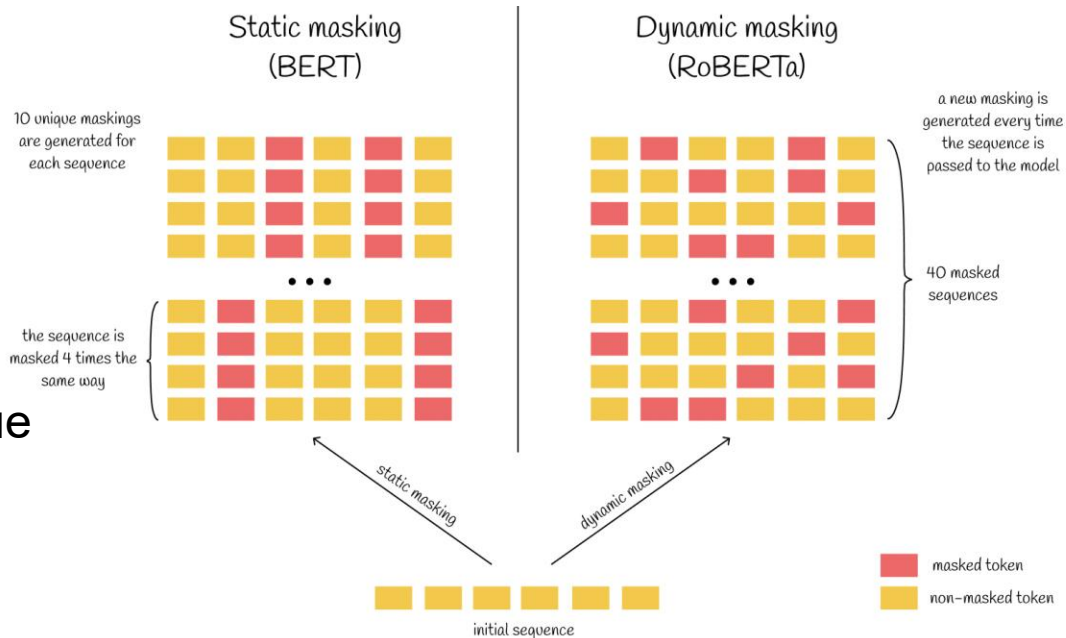
DYNAMIC MASKING

BERT:

- 40 epochs, but only 10 unique maskings for each sequence

RoBERTa:

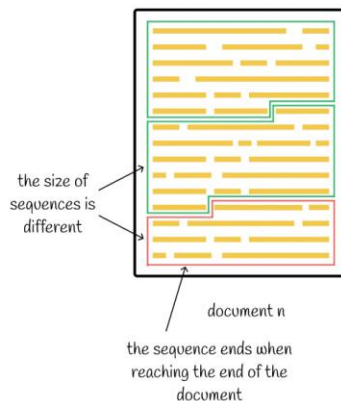
- 40 epochs with different masking each time



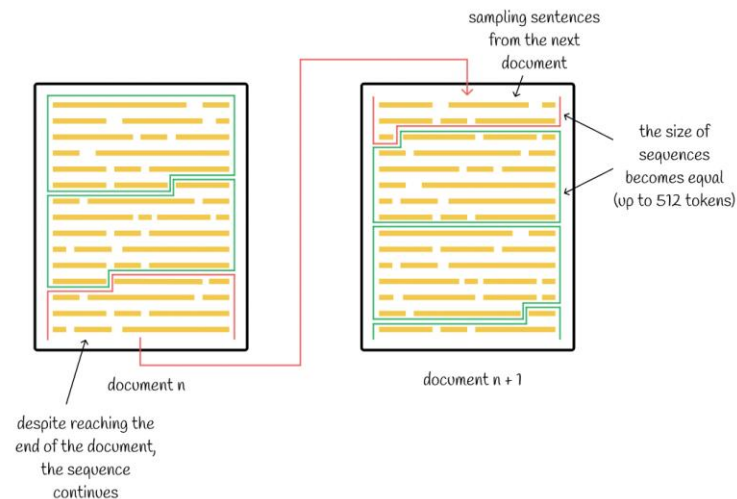
NEXT SENTENCE PREDICTION

- The next sentence prediction task was removed completely, because it turned out not to help with performance
- Passing sequences with several sentences worked better than passing single sentences

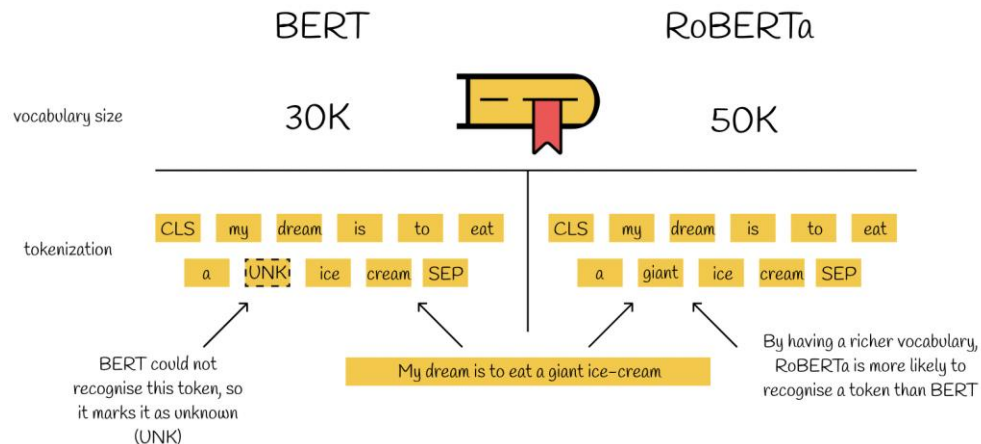
Sampling from a single document



Sampling from multiple documents (used in RoBERTa)



BYTE-BASED BPE

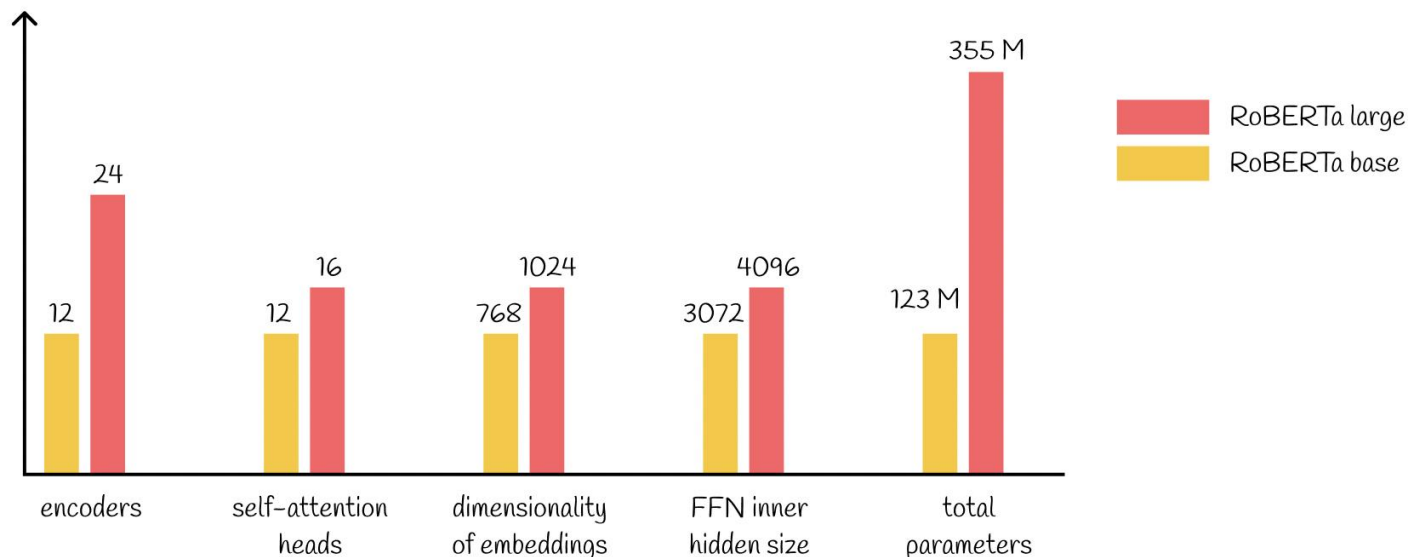


- Increased vocabulary size: 30k → 50k
- Bytes instead of Unicode characters as the base → every byte has an embedding → far fewer UNKs

PRETRAINING ROBERTA

- Much larger data size: 160GB total compared to 16GB for BERT
 - BookCorpus and English Wikipedia (16 GB, same as original BERT)
 - CC-News, from the English portion of CommonCrawl News (76 GB)
 - OpenWebText, web content extracted from URLs shared on reddit (38 GB)
 - Stories, a subset of CommonCrawl filtered to match a story-like format (31 GB)
- Longer training: 500k training steps compared to 100k for BERT

ROBERTA BASE VS ROBERTA LARGE



- Again, a base and a large version
- The only reason to use RoBERTa base would be constraints on the available compute

ROBERTA PERFORMANCE COMPARED TO BERT

	MNLI	QNLI	QQP	RTE	SST	MRPC	CoLA	STS	WNLI	Avg
<i>Single-task single models on dev</i>										
BERT _{LARGE}	86.6/-	92.3	91.3	70.4	93.2	88.0	60.6	90.0	-	-
XLNet _{LARGE}	89.8/-	93.9	91.8	83.8	95.6	89.2	63.6	91.8	-	-
RoBERTa	90.2/90.2	94.7	92.2	86.6	96.4	90.9	68.0	92.4	91.3	-

→ Better performance across the board (you don't need to know these individual datasets/tasks)

QUESTIONS?

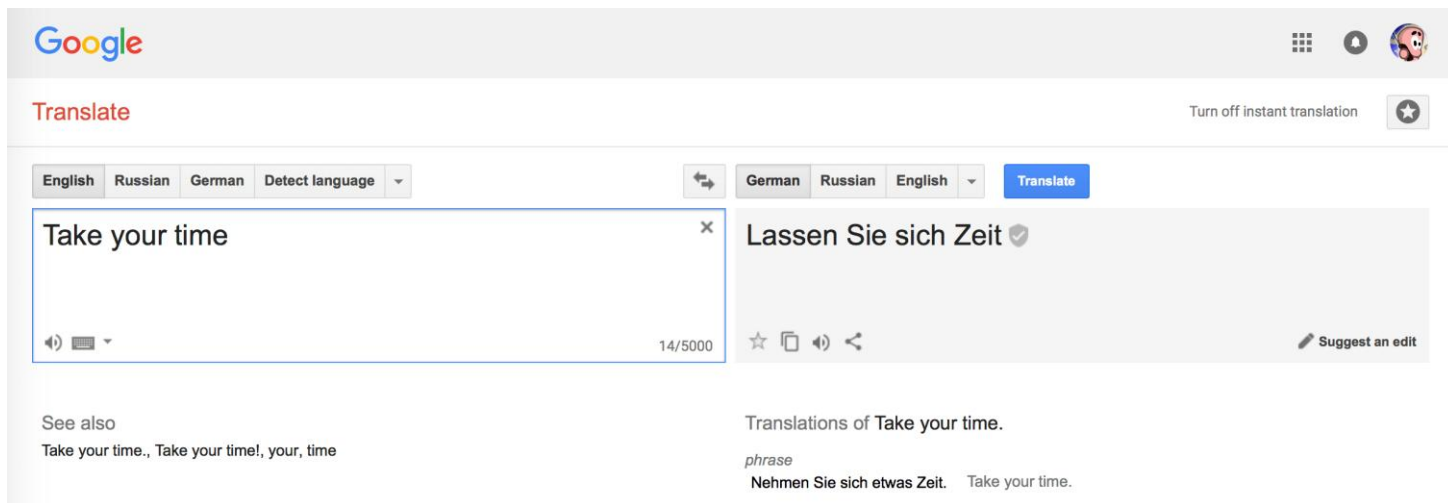
MULTILINGUAL BERT

MULTILINGUAL BERT

- Google released (without a paper) a version of BERT trained on data from 104 languages, specifically the 104 largest wikipedias at the time
- Important: The data was not parallel, there was no special signal or training, they just trained a BERT on the concatenation of all this data
- Multilingual BERT (mBERT) exhibited *cross-lingual transfer* capabilities: the ability to apply something it learned in one language in a different language

MACHINE TRANSLATION

The task of translating a (word, sentence, paragraph, document) from one language into another



EXAMPLE FOR EMERGENT MULTILINGUALITY IN BERT

- Sample 5k pairs from a machine-translation dataset
- Feed each sentence through mBERT
- Extract the hidden layer representation of [CLS] and [SEP] at each layer and average them to get a single vector representation for each sentence
- How often is the correct translation also the nearest neighbour in the resulting vector space?

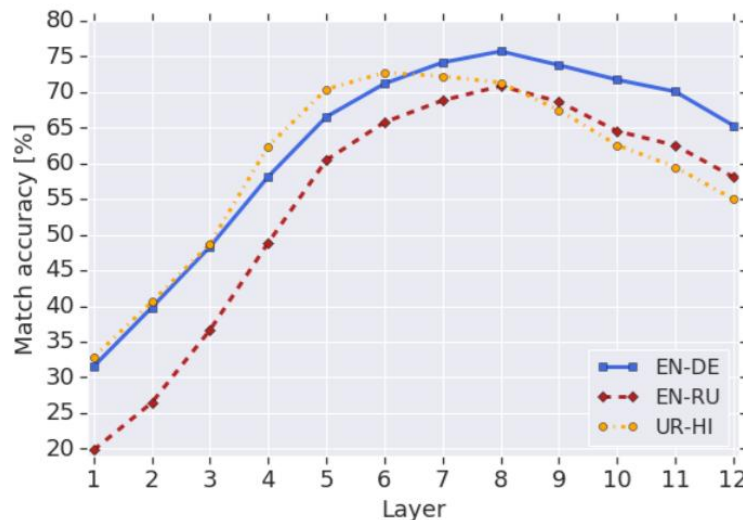


Figure 3: Accuracy of nearest neighbor translation for EN-DE, EN-RU, and HI-UR.

WHY DOES THIS WORK SO WELL, OR AT ALL?

- Multilingual BERT was never trained on machine translation or any other task with parallel data → it was never told which words in one language mean the same as other words in other languages
- And yet, mBERT has a shared representation space where the same meanings are close together, independent of the language!
- Hypothesis: having word pieces used in all languages (numbers, URLs, etc) which have to be mapped to a shared space forces the co-occurring pieces to also be mapped to a shared space, thus spreading the effect to other word pieces, until different languages are close to a shared space
- But really, we don't know!

QUESTIONS?

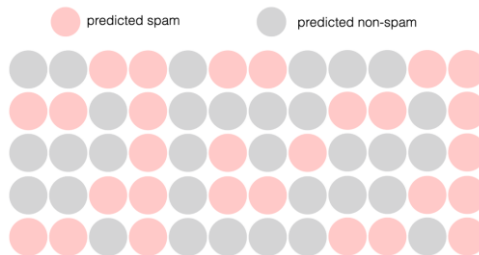
EVALUATION

ACCURACY

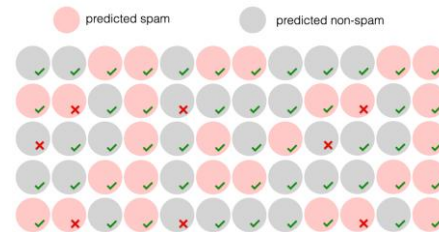
Accuracy answers the question: how often is the model right?

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$

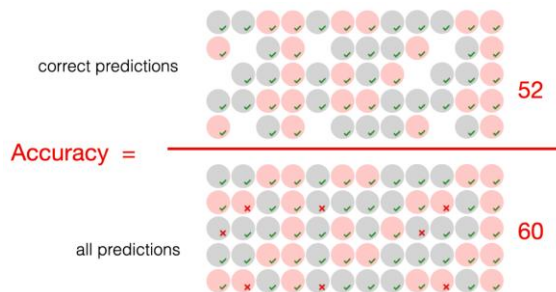
So for spam detection, for example:



And then count which predictions were wrong, using the gold labels:

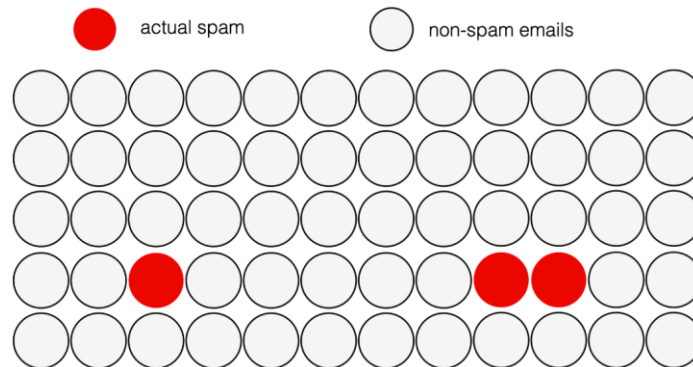


So the accuracy is

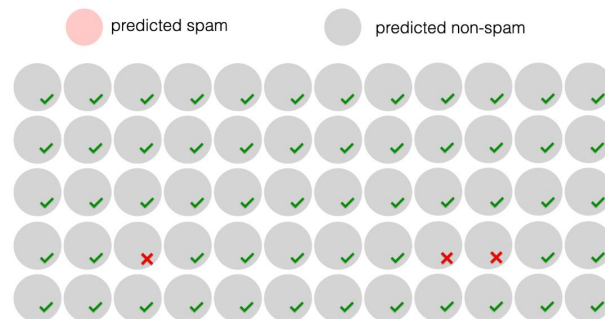


ACCURACY WHEN CLASSES ARE IMBALANCED

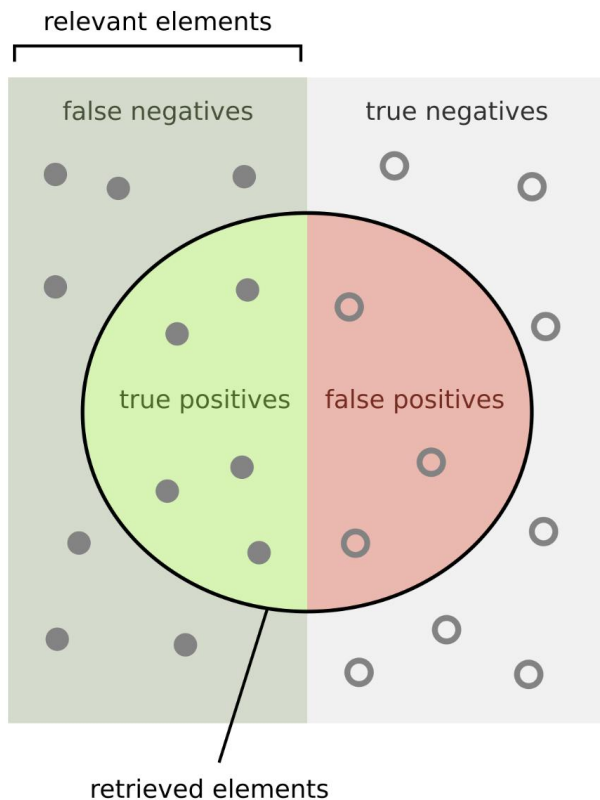
If very few emails (5%) are spam:



Then for a model, a simple way to reach almost perfect accuracy would be to never predict spam, for 95% accuracy:



PRECISION AND RECALL



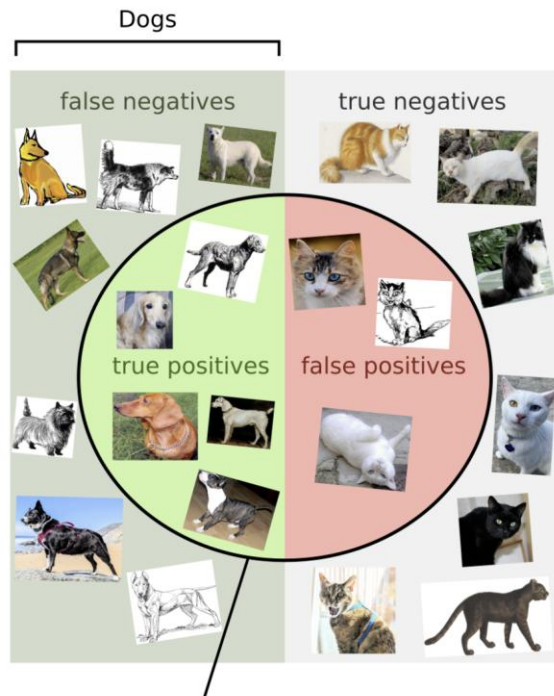
How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

PRECISION AND RECALL - EXAMPLE



images classified as having dogs

$$\text{Precision} = \frac{5 \text{ true pos.}}{8 \text{ total pos.}}$$

$$\text{Recall} = \frac{5 \text{ true pos.}}{12 \text{ total dogs}}$$

$$\text{Accuracy} = \frac{5 \text{ true pos.} + 7 \text{ true neg.}}{22 \text{ total images}}$$

CONFUSION MATRIX

		Predicted condition	
		Predicted positive	Predicted negative
Actual condition	Total population = P + N		
	Real Positive (P) ^[a]	True positive (TP), hit ^[b]	False negative (FN), miss, underestimation
	Real Negative (N) ^[d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection ^[e]

Useful to print out when doing classification, to see what the specific problems are

F1-SCORE

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2TP}{2TP + FP + FN}$$

- F1-Score is the harmonic mean between precision and recall
- Called F1-score because it's the special case of $\beta=1$ of the F-score:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}$$

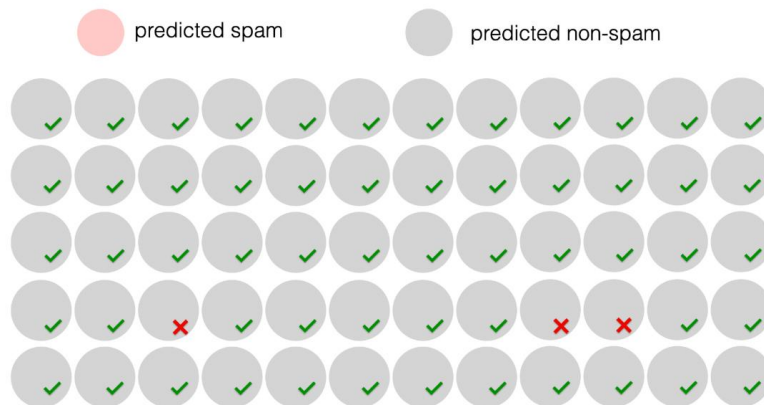
But other values of β are used very rarely in ML

BACK TO SPAM EMAILS

What are precision and recall if we always predict “no spam”?

Precision: 0 emails were labelled as spam, and out of these 0 detected positives, none were true positives $\rightarrow$ Precision is $0/0 = 1$

Recall: out of 3 spam emails that were to be found, none have been found $\rightarrow$ Recall is $0/3 = 0$

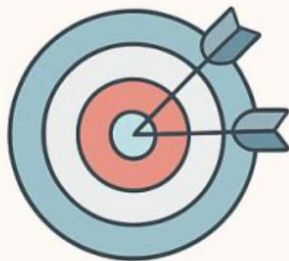


WHEN TO USE WHAT



ACCURACY

Use when
classes are
balanced



PRECISION

Use when
false positives
are costly



RECALL

Use when
false negatives
are costly

QUESTIONS?